

SHAY MORDECHAI

Vulnerability Researcher · Android Internals · Offensive Security Researcher

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SUMMARY

Offensive Security Researcher specializing in Android Internals and Native-level vulnerability discovery. Research experience includes Android Binder IPC, malware execution paths, browser runtimes (V8), compiler behavior analysis, and protocol implementation. Findings reported to CERT-IL and published through technical research articles and writeups.

TECHNICAL SKILLS

Reverse Engineering · Android Internals · Linux Internals · Frida (DBI) · IDA Pro · JADX · Fuzzing · Native-Hook · Binder IPC · Memory Models (V8/JIT) · Threat Modeling) · **Languages:** Python · C/C++ · Java · Assembly (x86)

PROJECTS & RESEARCH

Detailed write-ups at [shay-mordechai.github.io](https://github.com/shay-mordechai)

Android Internals & IPC Exploitation (Digital Whisper) | [Link](#)

- Researching Android system service communication by implementing low-level Binder IPC clients in C/C++ to analyze transaction validation between the WindowManagerService and SurfaceFlinger.

Android Malware Bootstrap Analysis (Digital Whisper) | [Link](#)

- Analyzed Anatsa banking trojan execution flow before UI initialization. Traced ContentProvider, attachBaseContext, LoadedApk, and class loading paths using Frida and runtime instrumentation. Documented early-stage application execution vectors and framework initialization behavior.

Compiler Security Research - SBCL Runtime (CERT-IL #11265) | [Link](#)

- Conducted semantic fuzzing research on Common Lisp macro expansion and compilation behavior. Identified three compile-time mechanisms challenging common assumptions regarding lexical isolation. Findings disclosed to CERT-IL and MITRE for evaluation.

PROFESSIONAL EXPERIENCE

Independent Security Research | Jul 2025 – Present

- Android Systems:** Leading architectural research on Android's UI/Graphics pipeline. Mapping privilege transitions (Bootstrap to App Context) and developing custom native IPC clients (C/C++) to expose Binder/SurfaceFlinger attack surfaces.
- Compiler & Infrastructure Security:** Conducted semantic fuzzing on SBCL runtimes (discovered 3 bypass vectors, MITRE/CERT-IL #11265) and designed Rust-based Envoy/WASM mitigations for financial API environments.
- Vulnerability Disclosure:** Managed multi-vendor coordination (Palo Alto Networks, TP-Link) for architectural security mitigations.

Application Security Analyst | Israel Tax Authority | Aug 2024 – Jul 2025

- Led enterprise SAST/SCA operations (Checkmarx) for production financial systems, conducting deep manual Code Reviews to validate complex Web/API attack vectors and analyze underlying exploitation mechanics.
- Spearheaded threat modeling and design-stage execution flow analysis for a secure GCP-based file ingestion cluster, eliminating upload flow abuse vectors and token handling vulnerabilities.
- Demonstrated practical mastery of the OWASP Top 10 by mentoring R&D teams; translated complex vulnerability findings into concrete remediation strategies and guided developers through secure coding practices within the SDLC.

EDUCATION

B.Sc. Computer Science | Jerusalem College of Technology | 2021 – 2025 · GPA: 89

Core Focus: Reverse Engineering (93) · Advanced Networks (97) · Advanced OOP (Java, C++)